YUANBO PANG

UC Berkeley EECS | Large-Scale Measurement & Data Systems
yuanbopang@berkeley.edu · (+1) 408-460-7526 · yuanbopang.pro · GitHub · LinkedIn

Education

University of California, Berkeley — B.S. EECS, GPA 3.762/4.0 (in-major 3.811) *Expected Dec 2026*
Coursework: Efficient Algorithms (CS 170), Discrete Math & Probability (CS 70), Machine Learning (Stanford CS 229), Data Structures (CS 61B)

Inspire Fellowship, rednote (Xiaohongshu) — $100K, one of 6 selected worldwide *2026*

Research Experience

Research Assistant, Berkeley RDI / BAIR (Prof. Dawn Song) *Oct 2025 – Present*
Agents' Last Exam (ALE) — co-first author arXiv:2606.05405

- Co-first author of ALE, a 1,000+ task benchmark measuring frontier AI agents on long-horizon real-world work; #1 Paper of the Week on Hugging Face. Led the task pipeline and the automated scoring system.
- Built cross-domain programmatic evaluators (hash, tabular, geometric, visual, behavioral) that score agent outputs with no hand labels — signals clean enough to double as RL training rewards; designed the checks that keep graders trustworthy as models and tasks drift.
- First author of **ALE-Replica** (under review, CVPR 2027): agents rebuild professional software from a few visible examples, scored on **held-out hidden suites** against real-software oracles (e.g. 3 visible / 47 hidden per operation); negative controls force shortcut solutions to zero. Collection: **130K+ cases across 76 applications**; pilot: 22 bundles, ∼17K cases, frontier agent at 34.8%.

Robotics Research, Prof. Masayoshi Tomizuka's Lab, UC Berkeley *2026 – Present*
roboqa — dataset-quality measurement at scale, advised by Zhuo Xu

- Built a benchmark that prices messy real-world data by what it teaches: health, diversity, and trainability scored separately (never averaged — units differ), on 338K episodes / 829 h of egocentric manipulation data.
- Versioned measurement limits and full provenance on every published score (data fingerprint, measure versions, judge, commit); domain-appropriate tolerances, with stochasticity recorded rather than hidden.
- Train and post-train VLA models on 500 h of curated data and score the resulting policies in simulation — dataset quality read as downstream model performance.

URAP Research Assistant, Prof. Avideh Zakhor, UC Berkeley EECS *Spring 2026 – Present*

- UAV-to-UAV detection and multi-object tracking from a moving camera (YOLO, optical flow, SAM2-style tracking); targeting a CVPR workshop.

Equile — The Dry Lab for AI Scientists (independent side project) *Apr 2026 – Oct 2026*

- Built reproducible computing environments where research agents run computational studies, each result carrying a provenance record; isolated sandboxes with multi-tenant packing and job scheduling (atomic claims, lease reclaim).

Publications

Agents' Last Exam — co-first author; under review, NeurIPS 2026; HF #1 of the week. **ALE-Replica** — first author; under review, CVPR 2027. **Agent's GameDev Exam** — 2nd author; under review, ICLR 2027. **ScienceIDE** (arXiv:2609.19134) — HF #2 of the day.

Skills

Core: Python (primary), PyTorch, C++, SQL, metric & evaluator design, large-scale data pipelines
Systems: Docker, GCP, Modal, Postgres, Linux; sandboxed execution and job scheduling at scale